Week 2

Case Study 1

Data for Questions 1 to 5

Clinic Follow-up Priority Scoring

A clinic uses a simple model to assign a follow-up priority score after remote consultation. The model does not make a diagnosis; it only produces a probability-like score for staff review. Each patient record is represented by two normalized inputs:

$$x_1 : \text{symptom persistence score}, \quad x_2 : \text{abnormal reading score}.$$

The model computes

$$z = 3x_1 + 2x_2 - 2, \qquad \hat{p} = \sigma(z) = \frac{1}{1 + e^{-z}}.$$

A record is sent for priority follow-up when $\hat{p} \geq 0.5$. For the sigmoid function, $\sigma(0) = 0.5$, positive z gives an output above 0.5, and negative z gives an output below 0.5.

Question 1: Why is the sigmoid function suitable for this priority score?

A) It maps a real-valued input to a value between 0 and 1.

B) It removes the need to compute any weighted input value.

C) It always gives a hard class label before thresholding.

D) It forces both input features to have identical weights.

Correct Answer: A

Explanation: A sigmoid maps any real number to a smooth output in $(0, 1)$, which can be used as a probability-like score.

Question 2: For $x_1 = 0.4$ and $x_2 = 0.6$, what is the value of z?

A) 0.2

B) 0.6

C) 0.4

D) 0.8

Correct Answer: C

Explanation: $z = 3(0.4) + 2(0.6) - 2 = 1.2 + 1.2 - 2 = 0.4$.

Question 3: The rule $\hat{p} \geq 0.5$ is equivalent to which condition?

A) $z < 0$

B) $z \geq 0$

C) $z > 1$

D) $z \leq -1$

Correct Answer: B

Explanation: Since $\sigma(0) = 0.5$ and sigmoid is increasing, $\hat{p} \geq 0.5$ exactly when $z \geq 0$.

Question 4: Which input is not sent for priority follow-up?

A) $x_1 = 0.8,\ x_2 = 0.1$

B) $x_1 = 0.2,\ x_2 = 0.8$

C) $x_1 = 0.5,\ x_2 = 0.4$

D) $x_1 = 0.1,\ x_2 = 0.4$

Correct Answer: D

Explanation: For $x_1 = 0.1,\ x_2 = 0.4$, $z = 0.3 + 0.8 - 2 = -0.9$, so $\hat{p} < 0.5$.

Question 5: If the clinic wants fewer records to satisfy $\hat{p} \geq 0.5$, what should happen to the bias term -2?

A) It should become more negative.

B) It should be removed from the model entirely.

C) It should be made equal to $x_1 + x_2$.

D) It should be increased to make every score larger.

Correct Answer: A

Explanation: Making the bias more negative lowers z for the same inputs, so fewer records reach the decision threshold.

Case Study 2

Data for Questions 6 to 10

Delivery Time Prediction Review

A logistics team trains a regression model to predict delivery delay in minutes. For four validation orders, the actual delay y and predicted delay $\hat{y}$ are:

Order	y	$\hat{y}$
1	12	10
2	20	24
3	15	13
4	18	21

The team uses squared error

$$L = (y - \hat{y})^2, \qquad \text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

For gradient-based updates with respect to the prediction,

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y).$$

Question 6: What is the squared error for Order 2?

A) 4

B) 16

C) 9

D) 25

Correct Answer: B

Explanation: For Order 2, the error is $20 - 24 = -4$, so the squared error is 16.

Question 7: What is the mean squared error over the four orders?

A) 7.25

B) 8.00

C) 9.50

D) 8.25

Correct Answer: D

Explanation: The squared errors are $4, 16, 4, 9$. Their sum is 33, so the MSE is $33/4 = 8.25$.

Question 8: Which order contributes the largest squared error?

A) Order 1

B) Order 3

C) Order 2

D) Order 4

Correct Answer: C

Explanation: The largest squared error is 16, which occurs for Order 2.

Question 9: If $y = 20$ and $\hat{y} = 24$, what is $\frac{\partial L}{\partial \hat{y}}$?

A) 8

B) -8

C) 4

D) -4

Correct Answer: A

Explanation: $\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) = 2(24 - 20) = 8$.

Question 10: A second model predicts $11, 21, 14, 19$. Which model has lower MSE?

A) The original model, because its MSE is 8.25.

B) The second model, because its MSE is 1.00.

C) Both models, because their errors are identical.

D) Neither model, because MSE cannot be computed here.

Correct Answer: B

Explanation: The second model has squared errors $1, 1, 1, 1$, so its MSE is 1.00.

Case Study 3

Data for Questions 11 to 15

Machine Energy Use Prediction

A facilities team monitors configurable machines in a shared workshop where different operating choices lead to different electricity usage patterns. Before deploying a larger model, the team studies one small feedforward network so that engineers can trace every intermediate value by hand. The system uses two real-valued inputs:

$$x_1 : \text{duration score}, \quad x_2 : \text{load score}.$$

The team uses a small feedforward network:

$$h_1 = \sigma(x_1 + x_2 - 1), \quad h_2 = \sigma(2x_1 - x_2),$$

$$\hat{y} = \sigma(3h_1 + 2h_2 - 2).$$

The hidden layer transforms the original features before the final output is computed. This lets the team discuss the order of computation, the role of nonlinear hidden units, and why the final sigmoid produces a bounded score rather than an unrestricted energy value.

Question 11: Which quantities are computed first in the feedforward direction?

A) The final loss value

B) The parameter update

C) The hidden units h_1 and h_2

D) The target label chosen by the model

Correct Answer: C

Explanation: In a feedforward network, the input layer is transformed into hidden unit values before computing the output.

Question 12: For $x_1 = 1$ and $x_2 = 1$, what is the input to the sigmoid in h_1?

A) 1

B) 0

C) 2

D) 3

Correct Answer: A

Explanation: The input to h_1 is $x_1 + x_2 - 1 = 1 + 1 - 1 = 1$.

Question 13: For $x_1 = 1$ and $x_2 = 0$, what is the input to the sigmoid in h_2?

A) -2

B) -1

C) 1

D) 2

Correct Answer: D

Explanation: The input to h_2 is $2x_1 - x_2 = 2(1) - 0 = 2$.

Question 14: Why can sigmoid hidden units introduce nonlinearity?

A) They copy inputs without changing any value.

B) They apply a nonlinear function to weighted input combinations.

C) They remove all weights from the network.

D) They force the output to be exactly linear for all possible inputs.

Correct Answer: B

Explanation: A sigmoid is nonlinear, so applying it in hidden units lets the network represent nonlinear transformations.

Question 15: What is the final output range of $\hat{y}$?

A) Any real number from $-\infty$ to ∞

B) Only the two values -1 and 1

C) A value between 0 and 1

D) A value greater than 1 for all inputs

Correct Answer: C

Explanation: The final output applies a sigmoid, so $\hat{y}$ lies in $(0, 1)$.

Case Study 4

Data for Questions 16 to 20

Warehouse Routing Time Analysis

A warehouse studies how a single routing coefficient affects the predicted travel time of a cart inside a storage aisle. The operations team first analyzes one clean training example so that the direction of a gradient descent step is easy to verify before the method is used on many routes. The one-parameter model is

$$\hat{y} = wx.$$

For one training example, $x = 3$ and $y = 15$. The squared error loss is

$$L(w) = (15 - 3w)^2.$$

The team studies this loss before using gradient descent on a larger training set. In this simplified setting, the parameter w controls how strongly the route length x affects the predicted time, and the squared loss measures how far the prediction is from the observed routing time.

Question 16: What is $L(w)$ when $w = 4$?

A) 6

B) 4

C) 12

D) 9

Correct Answer: D

Explanation: When $w = 4$, $\hat{y} = 3(4) = 12$, so $L = (15 - 12)^2 = 9$.

Question 17: Which value of w gives zero loss?

A) $w = 4$

B) $w = 5$

C) $w = 6$

D) $w = 3$

Correct Answer: B

Explanation: Zero loss requires $15 - 3w = 0$, so $w = 5$.

Question 18: What is $\frac{dL}{dw}$ at $w = 4$?

A) -18

B) -6

C) 18.0

D) 6

Correct Answer: A

Explanation: $L = (15 - 3w)^2$, so $\frac{dL}{dw} = -6(15 - 3w)$. At $w = 4$, this is -18.

Question 19: If gradient descent uses $w \leftarrow w - \eta\frac{dL}{dw}$ with $\eta = 0.1$, what is the next value from $w = 4$?

A) 3.2

B) 4.6

C) 5.8

D) 2.2

Correct Answer: C

Explanation: The update is $w = 4 - 0.1(-18) = 5.8$.

Question 20: What does the gradient indicate in this example?

A) It gives the class label directly without a model.

B) It removes the input feature from the prediction.

C) It makes the squared error impossible to compute.

D) It shows how the loss changes as w changes.

Correct Answer: D

Explanation: The gradient describes the local rate of change of the loss with respect to the parameter.